

COURSE TITLE
ENGINEERING ECONOMIC
ANALYSIS

Chapter 8 -INFLATION

**Concept of inflation and its causes, Measuring inflation,
Equivalence calculation under inflation, impact of inflation
on economic analysis**

Learning Objective

After studying this lecture students will be able to learn:

- Concept of inflation and terminologies**
- Causes of inflation**
- Measuring inflation**
- Equivalence calculation under inflation**
- Impact of inflation on economic analysis.**

Concept of inflation

- The value \$ 50 now cannot purchase the same amount of quantity of goods as \$50 used to buy the goods five or ten years ago.
- It means that the **value of currency is being decreased** as the time is passing by because of inflation.
- Inflation is the increase in the amount of money necessary to purchase the same amount of product or service over time.
[1]
- The cost of an item tends to increase over time, OR , the same dollar amount buys less of an item over time. [2]

Concept of inflation (Cont'd....)

- When the net demand for services and goods exceeds the total supply. Because of less supply, the net prices of these services/goods increases and this kind of situation is known as inflation.**
- Inflation produces effects on the health of an economy. For example- the uncertainty about the future behavior of “inflation rate” may restrict the investors to invest in the market.**

Concept of inflation (Cont'd....)

- **Deflation is the opposite of inflation** in that prices usually decreases over time, and hence, a specified dollar amount gains in purchasing power.
- Inflation is far more common than deflation in real world.
- Our focus will be restricted to accounting for inflation in economic analysis.
- If economic decisions are taken without considering the effect of inflation into account, most of them would become meaningless and as a result the organizations would end up with unpredictable return. [3]

Reasons of inflation

1. If the **production cost of various services and goods increases** then naturally the prices of the final products would also increase.
2. When industries and business houses **increase the total prices of their services and goods** in order to amplify their profit margins.
3. When a specific section of a mass industry increases the prices of its services and goods, because this step of a **particular section of a mass industry will produce considerable effects on various other sections of industry also**. For example- increase in the price of crude oil will spontaneously cause increase in the bus fares and airfares.

Causes of inflation

1. Demand pull inflation

‘ Too much money chases too few goods’

- Excessive spending power of consumers, sometimes obtained at the expense of savings, that pulls price up, called demand pull inflation [4]
- This occurs when there is excess aggregate demand in the economy or in the specific market or industry.
- It may be due to increase in population, decrease in taxes, increase in money supply etc.

Causes of inflation (Cont'd....)

2. Cost push inflation

- Increase in producer's costs that are passed along to customers, sometimes with disproportionate escalations that pushes price up, called cost push inflation. [4]**
- This occurs when the costs of production or operation are increasing.**
- It may be due to increase in the wages of labor, increase in price of the raw materials, increase in cost of imported materials etc.**

Causes of inflation (Cont'd....)

3. Increased money supply

- If the money supply increases faster than the rate of production, this could result in inflation, particularly demand-pull inflation because there will be too many dollars chasing too few products.**

[5]

4. Devaluation

- It is the policy of deliberate lowering the value of currency by a country.**

Causes of inflation (Cont'd....)

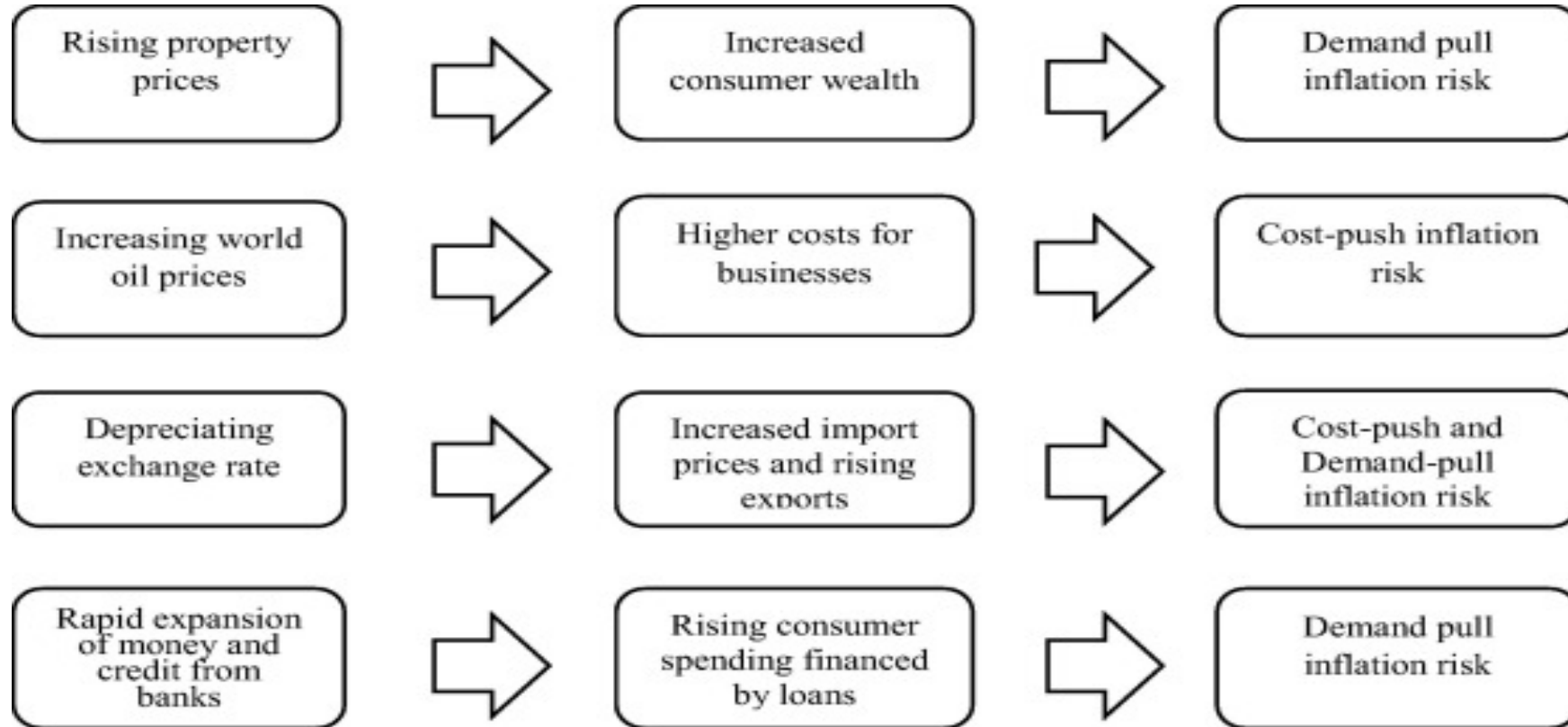


Figure : Causes of inflation [6]

Inflation Terminologies

1. Actual (current) dollars (A\$)

- Actual dollars are **estimates of future cash flow** for year 'n' that take into account any anticipated changes in amount **due to inflationary or deflationary effects**. [2]
- Usually these amounts are determined by applying an inflation rate to base year dollar estimates.

2. Constant (real) dollars (A' \$)

- Dollars expressed in terms of the same purchasing relative to a particular time.
- It means that the money have the constant purchasing power independent of passage of time. Does not consider inflationary and deflationary effects.

Inflation Terminologies (Cont'd.....)

3. Market interest rate (i)

- This rate takes into account the **combined effect** of the earning value of capital (earning power) and any anticipated inflation or deflation (purchasing power)

4. Inflation free interest rate (i')

- This rate is an estimate of the **true earning power** of the money when the **inflation effect have been removed**. This rate is commonly known as real interest rate.

5. Inflation rate (f)

- This is the measure of the **rate of change in the value of money**.

Measuring Inflation

- Generally inflation is measured in percentage term which is simply obtained by calculating the percentage change in current price index over the previous year.
- The most commonly price index used for measuring inflation are
 - (a) Consumer price index (CPI) and
 - (b) Producer price index (PPI)

Measuring Inflation (Cont'd....)

(a) Consumer price index (CPI)

□ The CPI measures **average change in the prices paid** for food, shelter, medical care, transportation and other goods and services used by individual or families.

(b) Producer price index (PPI)

□ The PPI is good measure of the **industrial price increases**.

□ It is a measure of changes in prices charged by manufacturers and wholesalers for products.

When performing the engineering economic analysis, the appropriate price indexes must be selected to estimate the price increase of raw materials, finished products and operating costs.

Measuring Inflation (Cont'd....)

Let's suppose we want to calculate the average inflation rate for a 2-year period: the first year's inflation rate is 4%, and the second year's rate is 8%, using a base price of \$ 100. [2]

- **Step 1:** To find the price at the end of the second year we use the process of compounding.

$$\begin{array}{c} \text{First year} \\ \longleftrightarrow \\ \$ 100 (1+0.04) (1+0.08) = \$112.32 \\ \longleftrightarrow \\ \text{Second Year} \end{array}$$

Measuring Inflation (Cont'd....)

Step 2: to find the average inflation rate (f), we establish the equivalence relation.

$$\text{\$ } 100 (1+f)^2 = \text{\$ } 112.32$$

$$\text{or, } \text{\$ } 100 (F/P, f, 2) = \text{\$ } 112.32$$

Solving for f yields,

$$f = 5.98\% = 6\% \text{ (approximately)}$$

If the inflation rate (f) averages 6% over the next 10 years,

$$= \$100 (1+0.06)^{10} = \$ 179.08 = \$ 180 \text{ (approximately)}$$

It can be concluded that the

% increase in the price of the product

$$= (\text{\$ } 180 - \text{\$ } 100) / \text{\$ } 100 = 80\%$$

Conversion of constant to actual dollar

Example

Consider the following net cash flow of a project in a constant dollar.

Period	Net Cash flow in Constant Dollar
0	-250,000
1	1,00,000
2	1,10,000
3	1,20,000
4	1,30,000
5	1,20,000

If the average inflation rate is projected as 5 % per year, convert the cash flow into actual dollars.

Conversion of constant to actual dollar

Example (Cont'd....)

Period	Cash flow in Constant Dollar	Conversion Factor = $(1+f)^n$	Cash flow in Actual Dollar
0	-250,000	$(1+0.05)^0$	-250,000
1	1,00,000	$(1+0.05)^1$	1,05,000
2	1,10,000	$(1+0.05)^2$	1,21,275
3	1,20,000	$(1+0.05)^3$	138,915
4	1,30,000	$(1+0.05)^4$	1,58,016
5	1,20,000	$(1+0.05)^5$	1,53,154

Equivalence Calculation Under Inflation

- In previous chapters, our equivalence analysis took into consideration changes in **earning power of money i.e. interest rate.**
- To factor in changes in purchasing power i.e. inflation, we use either **(1) constant dollar analysis (2) actual dollar analysis and (3) Mixed dollar analysis**

Procedure to adjust inflation under equivalence calculation[3]

- Estimate all the costs/returns associated with an investment proposal in terms of today's dollar.**
- Modify the costs/returns in step one using an assumed inflation rate so that at each future date they represent the costs/revenues at that day in terms of the dollars that must be expended or received at that time respectively.**
- As per our requirement, calculate either the annual equivalent amount or future amount or present amount of the cash flow resulting from step 2 by considering the time value of money.**

Case 1: If cash flow elements are estimated in Constant Dollar (Constant Dollar Analysis)

- Suppose that all cash flow elements are already given in constant dollar, and that we want to compute the equivalent PW of the constant dollar (A'_n) occurring in year 'n'.
- In the absence of inflationary effect, we use *inflation free interest rate (i')* to account only for the earning power of money.
- The equivalent present worth is obtained by

$$P_n = A'_n / (1+i')^n$$

Constant Dollar Analysis Numerical

Consider the constant dollar cash flows for a company as follows. If the manager want the company to earn a 12% inflation free rate of return before tax on any investment, what should be the present worth of this project?

Period	Net cash flow in Constant \$
0	- \$ 250,000
1	\$ 100,000
2	\$ 110,000
3	\$ 120,000
4	\$ 130,000
5	\$ 120,000

(Constant Dollar Analysis) Numerical

Since all values are in constant dollars, we can use the inflation free interest rate. We simply discount the dollar inflows at $i' = 12\%$

$$\text{PW (12\%)} = - \$ 250,000 + \$ 100,000 (P/A, 12\% , 5) + \$ 10,000 (P/G, 12\%, 4) + \$ 20,000 (P/F, 12\%, 5)$$

$$\text{PW (12\%)} = \$ 163,099$$

Since the equivalent net receipts exceeds the investment, the project can be justified before considering any tax effects.

Case 2: If cash flow elements are estimated in Actual Dollar (Actual Dollar Analysis)

□ In this method , all cash flow elements are estimated in actual dollars.

□ To find the equivalent present worth of this actual dollar amount (A_n) in year n , we may use either

(a) Deflation method or

(b) Adjusted-discount method.

Case 2: If cash flow elements are estimated in Actual Dollar (Actual Dollar Analysis) (Cont'd.....)

Deflation Method

- It requires **two steps** to convert actual dollars into equivalent present worth dollars.
- First, we convert actual dollars into equivalent constant dollars by **discounting by the general inflation rate** which removes the inflationary effect.
- Second, finding the equivalent present worth **using i' %**.

Actual Dollar Analysis (Deflation Method)

Numerical

The project is expected to generate the following cash flows in actual dollars

Period	Net cash flow in Actual \$
0	- \$ 75,000
1	\$ 32,000
2	\$ 35,700
3	\$ 32,800
4	\$ 29,000
5	\$ 58,000

- (a) What are the equivalent year 0 dollars (constant dollars), if the general inflation rate (f) is 5% per year?
- (b) Compute the present worth of these cash flows in constant dollars at $i' = 10\%$.

Deflation Method Numerical (Cont'd....)

The net cash flows in actual dollars can be converted to constant dollars **by deflating them by 5% yearly deflation factor**. The deflated or constant dollar cash flows can then be used to determine NPW at $i'\%$.

(a) Convert the actual dollars into constant dollars

Period	Cash flow in actual dollar	Multiplied by deflation factor	Cash flow in Constant dollar
0	- \$ 75,000	$(1+0.05)^{-0}$	- \$ 75,000
1	\$ 32,000	$(1+ 0.05)^{-1}$	\$ 30,476
2	\$ 35,700	$(1+0.05)^{-2}$	\$ 32,381
3	\$ 32,800	$(1+0.05)^{-3}$	\$ 28,334
4	\$ 29,000	$(1+0.05)^{-4}$	\$ 23,858
5	\$ 58,000	$(1+ 0.05)^{-5}$	\$ 45,445

Deflation Method Numerical (Cont'd....)

(b) Compute the equivalent present worth of constant dollar using $i' = 10\%$

Period	Cash flow in Constant dollar	Multiplied by discounting factor	Equivalent Present worth
0	- \$ 75,000	1	- \$ 75,000
1	\$ 30,476	$(1+ 0.10)^{-1}$	\$ 27,706
2	\$ 32,381	$(1+0.10)^{-2}$	\$ 26,761
3	\$ 28,334	$(1+0.10)^{-3}$	\$ 21,288
4	\$ 23,858	$(1+0.10)^{-4}$	\$ 16,295
5	\$ 45,445	$(1+ 0.10)^{-5}$	\$ 28,218
Total Present worth			\$ 45,268

Deflation Method Numerical (Cont'd....)

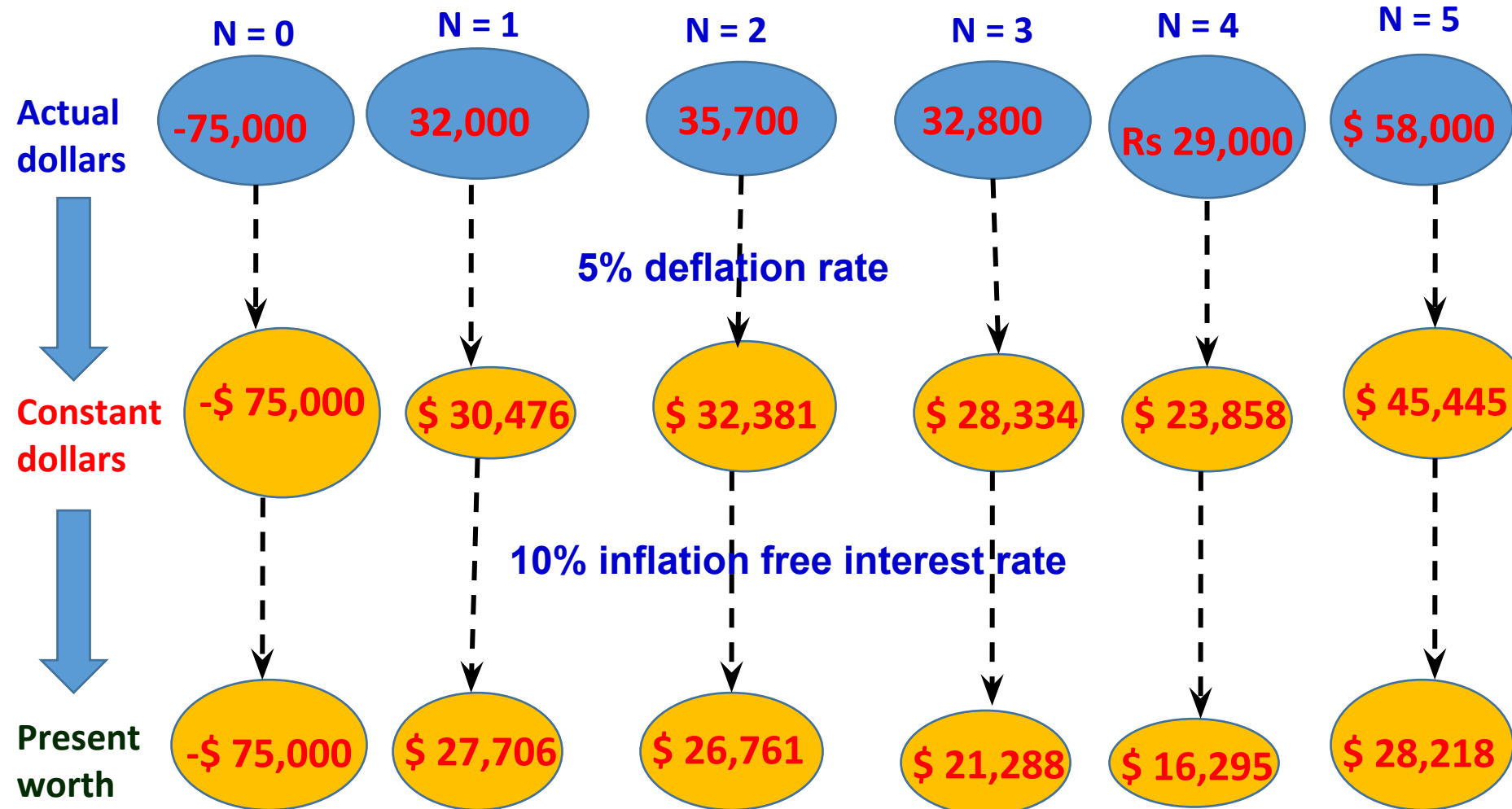


Figure : Converting actual dollar to constant dollar and then to equivalent present worth

Actual Dollar Analysis

Adjusted – discount method

The **two-step process** shown in previous example can be streamlined by the efficiency of the adjusted-discount method, **which performs deflation and discounting in one step.**

Mathematically,

we can combine this two step procedure into one step

$$P_n = \frac{\frac{A_n}{(1+f)^n}}{(1+i')^n}$$
$$P_n = \frac{A_n}{(1+f)^n (1+i')^n}$$

Adjusted – discount method (Cont'd...)

$$P_n = \frac{A_n}{\{(1+f)(1+i')\}^n} \dots\dots\dots (1)$$

Since the market interest rate (i) reflects both the earning power and the purchasing power, we have following relation relationship

$$P_n = \frac{A_n}{(1+i)^n} \dots\dots\dots (2)$$

The equivalent present worth values in equation (1) and (2) must be equal at year 0.

Adjusted – discount method (Cont'd...)

Therefore,

$$\frac{A_n}{\{(1+f)(1+i')\}^n} = \frac{A_n}{(1+i)^n}$$

This leads to the following relationship among f, i', i

$$(1+i) = (1+f)(1+i')$$

Simplifying the term yields ,

$$i = i' + f + i' * f$$

Adjusted Discount Method Numerical

Consider the cash flows in actual dollar as in previous example as follows. Compute the equivalent present worth of these flows using the adjusted discount method. $f = 5\%$, and $i' = 10\%$

Period	Net cash flow in Actual \$
0	- \$ 75,000
1	\$ 32,000
2	\$ 35,700
3	\$ 32,800
4	\$ 29,000
5	\$ 58,000

Adjusted Discount Method Numerical (Cont'd....)

First, determine the market interest rate (i)

$$i = i' + f + i' * f$$

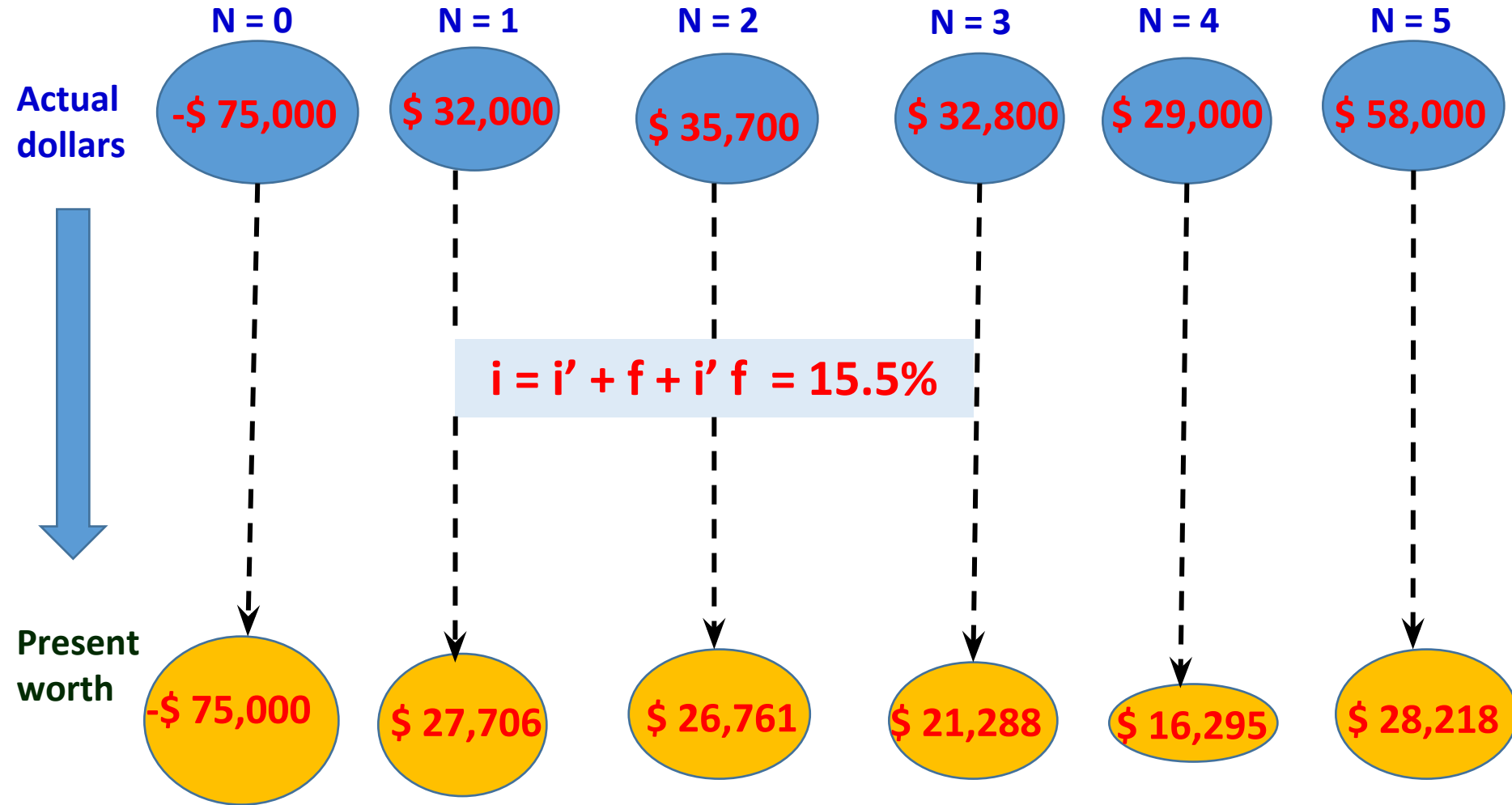
$$i = 0.10 + 0.05 + (0.10) (0.05)$$

$$i = 15.5 \%$$

Period	Cash flow in actual dollar	Multiplied by $(1+0.155)^{-n}$	Equivalent Present worth
0	- \$ 75,000	1	- \$ 75,000
1	\$ 32,000	$(1+ 0.155)^{-1}$	\$ 27,706
2	\$ 35,700	$(1+0.155)^{-2}$	\$ 26,761
3	\$ 32,800	$(1+0.155)^{-3}$	\$ 21,288
4	\$ 29,000	$(1+0.155)^{-4}$	\$ 16,295
5	\$ 58,000	$(1+ 0.155)^{-5}$	\$ 28,218
Total Present worth			\$ 45,268

The equivalent present worth that is obtained using adjusted – discount method ($i = 15.5\%$) is exactly same as the result obtained using the deflation method ($f=5\%$ and $i'=10\%$)

Adjusted Discount Method Numerical (Cont'd...)



Case3: Some of the cash flow elements are estimated in constant dollars and others are estimated in actual dollars.

Mixed Dollar Analysis

- **Convert all cash flow elements into the same dollar units (either constant or actual)**
- **If the cash flow is converted into actual dollars, the market interest rate is (i) should be used in calculating the equivalence value.**
- **If the cash flow is converted in terms of constant dollars, the inflation free interest rate (i') should be used.**

Impact of Inflation in engineering economic analysis

- The purchasing power of money never remains stationary or still, which indicates the importance of inflation in the economic world.**
- When inflation is modest, 2 to 4 percent per year, it is generally ignored in economic evaluations of proposals.**
- It is argued that all proposals are affected similarly by price changes and that there are too little differences between current and future costs to influence the order of preference.**
- When inflation is high and some goods and services escalate much more rapidly than others and there is a chance of project being suffered and cannot generate the income as expected.**

Impact of Inflation in engineering economic analysis (Cont'd....)

- Therefore it is necessary to make some consideration regarding the inflation while making the analysis of investment proposals.**
- There are two ways to make meaningful economic calculations when the value of the currency is changing [7]**
 - (a) Convert the amounts that occur in the different time periods into amounts that make the currencies have the same value before time value calculations are made, or**
 - (b) change the interest rate that is used in the economic equations in such a way that it accounts for the different valued currencies as well as the time value of money.**

References

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- [2] *Contemporary Engineering Economics*, Chan S. Park, Second Edition, Addison-Wesley Publishing Company, 1997.
- [3] *Engineering Economics: R. Paneerselvam*, Eastern Economy Edition, Prentice Hall of India Private Limited, New Delhi, India, 2001.

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[4] *Engineering Economics*: James L. Riggs, David D. Bedworth and Sabah U. Randhawa, Fourth Edition, Tata McGraw Hill Education Private Limited, New Delhi, India, 2004.

[5] <https://www.businessinsider.com/personal-finance/causes-of-inflation>

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[6]

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[7] <http://engineering.utep.edu/enge/EE/14/02/1.htm> (Viewed November

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THANK YOU